Package ‘manMetaVAR’

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From Person-Specific Dynamics to Population Inference:
A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework
for Integrating Complex Multilevel Dynamic Models.
<doi:10.0000/0000000000>.

URL `https://github.com/jeksterslab/manMetaVAR`,
`https://jeksterslab.github.io/manMetaVAR/`,
`https://osf.io/uenr7/`, `https://doi.org/10.0000/0000000000`

BugReports `https://github.com/jeksterslab/manMetaVAR/issues`

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Contents

`Check` *Check Replication*

Description

Check Replication

Usage

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitDTVAR *Check Replication - FitDTVAR*

Description

Check Replication - FitDTVAR

Usage

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitDTVARK4 *Check Replication - FitDTVARK4*

Description

Check Replication - FitDTVARK4

Usage

```
CheckFitDTVARK4(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `CheckK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMetaVAR` *Check Replication - FitMetaVAR*

Description

Check Replication - FitMetaVAR

Usage

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMetaVARK4` *Check Replication - FitMetaVARK4*

Description

Check Replication - FitMetaVARK4

Usage

```
CheckFitMetaVARK4(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `CheckK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMplus` *Check Replication - FitMplus*

Description

Check Replication - FitMplus

Usage

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMplusK4 — *Check Replication - FitMplusK4*

Description

Check Replication - FitMplusK4

Usage

```
CheckFitMplusK4(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `CheckK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMplusK4Priors — *Check Replication - FitMplusK4 (User Defined Priors)*

Description

Check Replication - FitMplusK4 (User Defined Priors)

Usage

```
CheckFitMplusK4Priors(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `CheckK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMplusPriors` *Check Replication - FitMplus (User Defined Priors)*

Description

Check Replication - FitMplus (User Defined Priors)

Usage

```
CheckFitMplusPriors(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitNaive *Check Replication - FitNaive*

Description

Check Replication - FitNaive

Usage

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by `.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckK4 *Check Replication*

Description

Check Replication

Usage

```
CheckK4(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Compression Functions: `CompressK4()`

CompressK4 — *Compress Replication*

Description

Compress Replication

Usage

```
CompressK4(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Compression Functions: `Compress()`

diagnostics — *Bayesian Mplus Diagnostics Across Simulation Conditions*

Description

Parameter- and run-level Bayesian Mplus diagnostics for all simulation task IDs and both prior specifications.

Usage

```
data(diagnostics)
```

Format

A list with three elements:

parameters A data frame with one row per task ID, replication, prior condition, and Mplus parameter. It contains the simulation design variables, posterior summaries, R-hat, bulk and tail ESS, and MCSEs.

runs A data frame with one row per task ID, replication, and prior condition containing run-level diagnostics and elapsed time.

replications Number of simulation replications represented in each task/prior cell.

The object is generated by `.setup/data-process/data-process-z-diagnostics.R` from the task-specific outputs of `SumFitMplusDiagnostics()` and `SumFitMplusPriorsDiagnostics()`.

Author(s)

Ivan Jacob Agaloos Pesigan

`FigMetricHeatmap` *Results Heatmap*

Description

Plot results heatmap.

Usage

```
FigMetricHeatmap(results, metric_name, grey_scale = FALSE, values = FALSE)
```

Arguments

`results`	`results` data frame.
`metric_name`	Character string. `"coverage"` for coverage probability, `"abs_rel_bias"` for absolute value of the relative bias, `"rmse"` for RMSE, `"power"` for statistical power, and `"type1_error"` for the Type I error rate.
`grey_scale`	Logical. If `TRUE`, use a grey-scale fill suitable for black-and-white printing.
`values`	Logical. If `TRUE`, display values inside tiles.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigMplusDiagnosticsHeatmap()`, `FigMplusDiagnosticsOverview()`, `FigOverview()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigMetricHeatmap(
  results = results,
  metric_name = "coverage"
)

## End(Not run)
```

FigMplusDiagnosticsHeatmap

Bayesian Diagnostic Heatmaps Across Simulation Conditions

Description

Summarize parameter-level Mplus Bayesian diagnostics across all simulation task IDs and display them using the same design-cell-by-parameter layout as `FigMetricHeatmap()`. The function can display the two prior conditions directly or paired user-prior minus default-prior differences.

Usage

```
FigMplusDiagnosticsHeatmap(
  diagnostics,
  metric_name = "failure_rate",
  comparison = "difference",
  diagnostic = NULL,
  taskid = NULL,
  parm = NULL,
  center = "mean",
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  grey_scale = FALSE,
  values = FALSE
)
```

Arguments

`diagnostics`	An all-task diagnostic object containing a `parameters` data frame, such as the `diagnostics` object generated by `.setup/data-process/data-process-z-diagnostics.R`. A parameter-level data frame may also be supplied directly.
`metric_name`	Character string. Valid values are `"failure_rate"`, `"rhat"`, `"ess"`, and `"mcse"`.
`comparison`	Character string. Use `"levels"` to show default and user-specified priors or `"difference"` to show paired user-prior minus default-prior differences.

`diagnostic`	Optional diagnostic panel names to retain. If `NULL`, all panels for the selected metric are displayed.
`taskid`	Optional positive integer task IDs to include. The default is to summarize all task IDs.
`parm`	Parameter names or positive integer positions to include. If `NULL`, all k = 2 Mplus parameters are included.
`center`	Character string indicating whether continuous diagnostics are summarized using the `"mean"` or `"median"` across replications. Failure rates always use the mean.
`rhat_threshold`	Numeric R-hat threshold.
`ess_threshold`	Numeric bulk and tail ESS threshold.
`mcse_threshold`	Numeric MCSE threshold.
`relative_mcse`	Logical. If `TRUE`, Monte Carlo standard errors are divided by the posterior standard deviation before summarizing.
`grey_scale`	Logical. If `TRUE`, use a grey-scale fill.
`values`	Logical. If `TRUE`, print summarized values inside cells.

Details

Failure rates count nonfinite diagnostic values as failures. For `comparison = "difference"`, diagnostic values are paired by task ID, replication, parameter, and diagnostic before computing user-prior minus default-prior differences. Blue cells indicate improvement under user priors and red cells indicate deterioration.

Value

A `ggplot` object. Its `data` element contains the all-task summary used in the heatmap.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigMetricHeatmap()`, `FigMplusDiagnosticsOverview()`, `FigOverview()`

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

FigMplusDiagnosticsHeatmap(
  diagnostics = diagnostics,
  metric_name = "failure_rate",
  comparison = "difference"
)

FigMplusDiagnosticsHeatmap(
```

```
    diagnostics = diagnostics,
    metric_name = "rhat",
    comparison = "levels",
    values = TRUE
  )

## End(Not run)
```

FigMplusDiagnosticsOverview
Compact Bayesian Diagnostic Overview Across Simulation Conditions

---

Description

Summarize Mplus Bayesian diagnostic failures across all simulation task IDs at the Innovation, fixed-effect, and random-effect block level. This compact overview is intended for manuscript figures and reviewer-oriented prior sensitivity analyses. Use `FigMplusDiagnosticsHeatmap()` for parameter-level drill-down figures.

Usage

```
FigMplusDiagnosticsOverview(
  diagnostics,
  comparison = "difference",
  diagnostic = "Any diagnostic",
  unit = "replication",
  taskid = NULL,
  target = NULL,
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  grey_scale = FALSE,
  values = FALSE,
  label_threshold = 0.05
)
```

Arguments

`diagnostics`	An all-task diagnostics object containing a `parameters` data frame, or the parameter-level data frame itself.
`comparison`	Character string. Use `"levels"` to show diagnostic failure rates under each prior specification or `"difference"` to show paired user-prior minus default-prior differences.

diagnostic — Diagnostic failure panels to display. Available values are "Any diagnostic", "R-hat", "Bulk ESS", "Tail ESS", and "Relative MCSE" when relative_mcse = TRUE.

unit — Character string controlling the unit summarized within each design cell. "replication" classifies a replication as failed when any parameter in the selected parameter block fails. "parameter" averages failure indicators across parameter-replication pairs.

taskid — Optional positive integer task IDs to include.

target — Optional parameter blocks to include. Available values are "Innovation", "FE", and "RE".

rhat_threshold — Numeric R-hat threshold.

ess_threshold — Numeric bulk and tail ESS threshold.

mcse_threshold — Numeric MCSE threshold.

relative_mcse — Logical. If TRUE, Monte Carlo standard errors are divided by the posterior standard deviation before classifying failures.

grey_scale — Logical. If TRUE, use grey-scale fills.

values — Logical. If TRUE, print values inside sufficiently large cells.

label_threshold — Numeric value between zero and one. Labels with an absolute value smaller than this threshold are omitted. For level plots, the threshold applies directly to the failure rate.

Details

The default unit = "replication" answers a reviewer-friendly question: what proportion of simulation replications had at least one diagnostic problem within each parameter block? For comparison = "difference", the function pairs prior specifications within task ID and replication before calculating user-prior minus default-prior differences. Negative values indicate fewer failures under user-specified priors.

Value

A ggplot object. Its data element contains the block-level summaries used in the figure.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigMetricHeatmap(), FigMplusDiagnosticsHeatmap(), FigOverview()

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

FigMplusDiagnosticsOverview(
  diagnostics = diagnostics,
  comparison = "levels"
)

FigMplusDiagnosticsOverview(
  diagnostics = diagnostics,
  comparison = "difference",
  values = TRUE
)

## End(Not run)
```

`FigOverview` *Results Overview*

Description

Plot results overview.

Usage

```
FigOverview(results)
```

Arguments

`results`	`results` data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigMetricHeatmap()`, `FigMplusDiagnosticsHeatmap()`, `FigMplusDiagnosticsOverview()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigOverview(results)

## End(Not run)
```

FitDTVAR — *Fit the Model using the fitVARMxID Package*

Description

The function fits the model using the fitVARMxID package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVARK4(), FitMetaVAR(), FitMetaVARK4(), FitMplus(), FitMplusDiagnostics(), FitMplusK4(), FitMplusK4Diagnostics(), FitNaive(), FitNaiveK4(), MplusInput(), MplusInputK4()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitDTVARK4 — *Fit the Four-Variable Model using the fitVARMxID Package*

Description

The function fits the four-variable feasibility model using the fitVARMxID package.

Usage

```
FitDTVARK4(data, ncores = NULL, seed = NULL)
```

Arguments

`data`	R object. Output of the `GenDataK4()` function.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitMetaVARK4()`, `FitMplus()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
fit <- FitDTVARK4(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

`FitMetaVAR` *Multivariate Meta-Analysis using the metaDyn Package*

Description

The function performs multivariate meta-snalysis using the metaDyn package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVAR()` function.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVARK4()`, `FitMplus()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

FitMetaVARK4 — *Multivariate Meta-Analysis for the Four-Variable Model*

Description

The function performs multivariate meta-analysis for the four-variable feasibility model using the metaDyn package. The random-effects covariance matrix is diagonal, consistent with the population specification in `modelk4`.

Usage

```
FitMetaVARK4(fit, ncores = NULL, seed = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVARK4()` function.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVAR()`, `FitMplus()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
fit <- FitDTVARK4(data = data, seed = seed)
meta <- FitMetaVARK4(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
```

```
vcov(meta)

## End(Not run)
```

FitMplus *Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use `Mplus` default priors.
`wd`	Character string. Working directory.
`mplus_bin`	Character string. Path to Mplus binary. If `mplus_bin = NULL`, the function will try to find the appropriate binary.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVAR()`, `FitMetaVARK4()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
dsem <- FitMplus(data = data)
print(dsem)
summary(dsem)
coef(dsem)
vcov(dsem)
confint(dsem)
plot(dsem)
plot(dsem, what = "trace")

## End(Not run)
```

FitMplusDiagnostics — *Posterior Diagnostics for FitMplus*

Description

The function computes parameter-level posterior summaries and Markov chain Monte Carlo diagnostics from the posterior draws saved by `FitMplus()`.

Usage

```
FitMplusDiagnostics(object, burnin = NULL, level = 0.95)
```

Arguments

`object`	Object of class `manmetavar.mplus` returned by `FitMplus()`.
`burnin`	Integer indicating the number of initial draws to discard from each chain. If `burnin = NULL`, use `object$burnin`.
`level`	Numeric value indicating the credibility level.

Value

An object of class `manmetavar.mplus.diagnostics` containing a parameter-level diagnostics data frame and a run-level diagnostics data frame.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVAR()`, `FitMetaVARK4()`, `FitMplus()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data, seed = seed)
diagnostics <- FitMplusDiagnostics(fit)
diagnostics$parameters
diagnostics$run

## End(Not run)
```

FitMplusK4 *Fit the Four-Variable Model using Mplus*

Description

The function fits the four-variable feasibility model using Mplus.

Usage

```
FitMplusK4(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use `Mplus` default priors.
`wd`	Character string. Working directory.
`mplus_bin`	Character string. Path to Mplus binary. If `mplus_bin = NULL`, the function will try to find the appropriate binary.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitDTVARK4(), FitMetaVAR(), FitMetaVARK4(), FitMplus(), FitMplusDiagnostics(), FitMplusK4Diagnostics(), FitNaive(), FitNaiveK4(), MplusInput(), MplusInputK4()

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
dsem <- FitMplusK4(data = data)
print(dsem)
summary(dsem)
coef(dsem)
vcov(dsem)
confint(dsem)
plot(dsem)
plot(dsem, what = "trace")

## End(Not run)
```

FitMplusK4Diagnostics *Posterior Diagnostics for FitMplusK4*

Description

The function computes parameter-level posterior summaries and Markov chain Monte Carlo diagnostics from the posterior draws saved by `FitMplusK4()`.

Usage

```
FitMplusK4Diagnostics(object, burnin = NULL, level = 0.95)
```

Arguments

object	Object of class `manmetavar.mplus.k4` returned by `FitMplusK4()`.
burnin	Integer indicating the number of initial draws to discard from each chain. If `burnin = NULL`, use `object$burnin`.
level	Numeric value indicating the credibility level.

Value

An object of class `manmetavar.mplus.k4.diagnostics` containing a parameter-level diagnostics data frame and a run-level diagnostics data frame.

See Also

Other Model Fitting Functions: FitDTVAR(), FitDTVARK4(), FitMetaVAR(), FitMetaVARK4(), FitMplus(), FitMplusDiagnostics(), FitMplusK4(), FitNaive(), FitNaiveK4(), MplusInput(), MplusInputK4()

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
fit <- FitMplusK4(data = data, seed = seed)
diagnostics <- FitMplusK4Diagnostics(fit)
diagnostics$parameters
diagnostics$run
print(diagnostics)

## End(Not run)
```

FitNaive *Naive*

Description

The function performs the naive "fit-many-then-summarize".

Usage

```
FitNaive(fit, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitDTVARK4(), FitMetaVAR(), FitMetaVARK4(), FitMplus(), FitMplusDiagnostics(), FitMplusK4(), FitMplusK4Diagnostics(), FitNaiveK4(), MplusInput(), MplusInputK4()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
naive <- FitNaive(fit = fit, seed = seed)
summary(naive)
```

```
print(naive)
coef(naive)
vcov(naive)

## End(Not run)
```

FitNaiveK4 *Naive Estimation for the Four-Variable Model*

Description

The function performs the naive "fit-many-then-summarize" procedure for the four-variable feasibility model. The random-effects covariance matrix is diagonal, consistent with `modelk4`.

Usage

```
FitNaiveK4(fit, seed = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVAR()` function.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVAR()`, `FitMetaVARK4()`, `FitMplus()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `MplusInput()`, `MplusInputK4()`

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
fit <- FitDTVARK4(data = data, seed = seed)
naive <- FitNaiveK4(fit = fit, seed = seed)
summary(naive)
print(naive)
coef(naive)
vcov(naive)

## End(Not run)
```

GenData *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

`taskid`	Positive integer. Task ID.
`seed`	Integer. Random seed.

See Also

Other Data Generation Functions: `GenDataK4()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

GenDataK4 *Simulate Four-Variable Data*

Description

The function simulates data for the four-variable feasibility analysis using `simStateSpace::SimSSMVARIVary()`. The simulation condition is obtained from `taskid`, whereas the population model parameters are obtained from `modelk4`.

Usage

```
GenDataK4(taskid, seed = NULL)
```

Arguments

`taskid`	Positive integer. Task ID.
`seed`	Integer. Random seed.

See Also

Other Data Generation Functions: `GenData()`

Examples

```
## Not run:
seed <- 42
data <- GenDataK4(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

`manmetavar-data-methods`

Methods for Objects of Class `manmetavar.data`

Description

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

`x`	Object of class `manmetavar.data`.
`...`	additional arguments.
`object`	Object of class `manmetavar.data`.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-dtvar-methods
Methods for Objects of Class `manmetavar.dtvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvar`.

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
```

```
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

Arguments

object	Object of class manmetavar.dtvar.
mu	Logical. If mu = TRUE, include estimates of the mu vector, if available. If mu = FALSE, exclude estimates of the mu vector.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
ncores	Positive integer. Number of cores to use.
...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix.
x	Object of class manmetavar.dtvar.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-metavar-methods

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

This page documents the available methods for objects of class `manmetavar.naive`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

`object`	Object of class `manmetavar.naive`.
`...`	additional arguments.

robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available.
x	Object of class manmetavar.naive.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-mplus-k4-methods

Methods for Objects of Class manmetavar.mplus.k4

Description

This page documents the available methods for objects of class manmetavar.mplus.k4.

Usage

```
## S3 method for class 'manmetavar.mplus.k4'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus.k4'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus.k4'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus.k4'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus.k4'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

object	Object of class manmetavar.mplus.k4.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.
burnin	Integer indicating initial samples to discard. If burnin = NULL, use the burn-in stored in object$burnin.
...	additional arguments.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class manmetavar.mplus.k4.
what	Character string. If what = "posterior", return posterior distribution plots. If what = "trace", return trace plots. If what = "rhat", "ess", or "mcse", return the corresponding Bayesian diagnostic plot from FitMplusK4Diagnostics().
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-mplus-methods

Methods for Objects of Class manmetavar.mplus

Description

This page documents the available methods for objects of class manmetavar.mplus.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
```

```
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

object	Object of class manmetavar.mplus.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.
burnin	Integer indicating initial samples to discard. If burnin = NULL, use the burn-in stored in object$burnin.
...	additional arguments.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class manmetavar.mplus.
what	Character string. If what = "posterior", return posterior distribution plots. If what = "trace", return trace plots. If what = "rhat", "ess", or "mcse", return the corresponding Bayesian diagnostic plot from FitMplusDiagnostics().
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

---

manmetavar-naive-k4-methods

Methods for Objects of Class manmetavar.naive.k4

Description

This page documents the available methods for objects of class manmetavar.naive.k4.

Usage

```
## S3 method for class 'manmetavar.naive.k4'
coef(object, ...)

## S3 method for class 'manmetavar.naive.k4'
vcov(object, ...)

## S3 method for class 'manmetavar.naive.k4'
print(x, ...)

## S3 method for class 'manmetavar.naive.k4'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

`object`	Object of class `manmetavar.naive.k4`.
`...`	additional arguments.
`x`	Object of class `manmetavar.naive.k4`.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

`model` *Model Parameters*

Description

Model Parameters

Usage

```
data(model)
```

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector $\boldsymbol{\mu}$.

mu_sigma Covariance matrix of the parameter $\boldsymbol{\mu}$.

mu_sigma_l Cholesky factor of the covariance matrix of the parameter $\boldsymbol{\mu}$.

beta_mu Mean of lagged coefficients matrix $\boldsymbol{\beta}$.

beta_sigma Covariance matrix of the parameter $\text{vec}(\boldsymbol{\beta})$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\text{vec}(\boldsymbol{\beta})$.

psi Process noise covariance matrix $\boldsymbol{\Psi}$.

psi_l Cholesky factor of the process noise covariance matrix $\boldsymbol{\Psi}$.

psi_d_ldl `uc_d` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

psi_l_ldl `s_l` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

ma_fixed Vector of fixed effects $\boldsymbol{\theta} = [\boldsymbol{\mu}, \text{vec}(\boldsymbol{\beta})]'$

ma_random Matrix of random effects $\boldsymbol{\theta} = [\boldsymbol{\mu}, \text{vec}(\boldsymbol{\beta})]'$

ma_random_d_ldl `uc_d` of the LDL' decomposition of the random-effects covariance. See `fitVARMxID::LDL()`.

ma_random_l_ldl `s_l` of the LDL' decomposition of the random-effects covariance. See `fitVARMxID::LDL()`.

Author(s)

Ivan Jacob Agaloos Pesigan

`modelk4` *Four-Variable Model Parameters*

Description

Population model parameters for the four-variable VAR(1) simulation condition.

Usage

```
data(modelk4)
```

Format

A list with 15 elements:

k Number of variables, equal to 4.

mu_mu Mean of the set-point vector $\boldsymbol{\mu}$.

mu_sigma Covariance matrix of the parameter $\boldsymbol{\mu}$.

mu_sigma_l Cholesky factor of the covariance matrix of the parameter $\boldsymbol{\mu}$.

beta_mu Mean of the lagged-coefficient matrix $\boldsymbol{\beta}$.

beta_sigma Covariance matrix of the parameter $\text{vec}(\boldsymbol{\beta})$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\text{vec}(\boldsymbol{\beta})$.

psi Process-noise covariance matrix $\boldsymbol{\Psi}$.

psi_l Cholesky factor of the process-noise covariance matrix $\boldsymbol{\Psi}$.

psi_d_ldl `uc_d` of the LDL' decomposition of the process-noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

psi_l_ldl `s_l` of the LDL' decomposition of the process-noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

ma_fixed Vector of fixed effects $\boldsymbol{\theta} = [\boldsymbol{\mu}, \text{vec}(\boldsymbol{\beta})]'$.

ma_random Covariance matrix of the random effects in $\boldsymbol{\theta} = [\boldsymbol{\mu}, \mathrm{vec}\,(\boldsymbol{\beta})]'$.

ma_random_d_ldl `uc_d` of the LDL' decomposition of the random-effects covariance matrix. See `fitVARMxID::LDL()`.

ma_random_l_ldl `s_l` of the LDL' decomposition of the random-effects covariance matrix. See `fitVARMxID::LDL()`.

Author(s)

Ivan Jacob Agaloos Pesigan

`MplusInput` *Create Mplus Input File*

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors = TRUE,
  ncores = NULL,
  seed = NULL
)
```

Arguments

`fn_data`	Character string. Filename for data file.
`fn_posterior`	Character string. Filename for posterior output.
`fn_factorscores`	Character string. Filename for factor scores output.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use `Mplus` default priors.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitDTVARK4(), FitMetaVAR(), FitMetaVARK4(), FitMplus(), FitMplusDiagnostics(), FitMplusK4(), FitMplusK4Diagnostics(), FitNaive(), FitNaiveK4(), MplusInputK4()

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL,
    seed = 42
  )
)
```

MplusInputK4 *Create Mplus Input File for the Four-Variable Model*

Description

The function creates an Mplus input file for the four-variable feasibility model. The between-person random-effects covariance matrix is diagonal, consistent with modelk4.

Usage

```
MplusInputK4(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors = TRUE,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
default_priors	Logical. If `default_priors = TRUE`, use `Mplus` default priors.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitDTVARK4()`, `FitMetaVAR()`, `FitMetaVARK4()`, `FitMplus()`, `FitMplusDiagnostics()`, `FitMplusK4()`, `FitMplusK4Diagnostics()`, `FitNaive()`, `FitNaiveK4()`, `MplusInput()`

Examples

```
cat(
  MplusInputK4(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL,
    seed = 42
  )
)
```

---

params *Simulation Parameters*

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 12 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions. If `NA`, measurement occasions vary across IDs.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

Author(s)

Ivan Jacob Agaloos Pesigan

`population` *Calibrated Population Parameters*

Description

Population moments implied by the stationary transition-matrix generator. The moments of the set points are analytical. The moments of the transition matrices are calculated by Monte Carlo integration over the stationary region for each nonzero heterogeneity condition.

Usage

```
data(population)
```

Format

A list with two elements:

conditions Named list indexed by heterogeneity. Each element contains the calibrated means and covariance matrices, a named population parameter vector, and Monte Carlo calibration information.

calibration Metadata describing the generator, package version, random seed, population size, batch size, stationarity margin, and generation time.

Author(s)

Ivan Jacob Agaloos Pesigan

populationk4 *Calibrated Four-Variable Population Parameters*

Description

Population moments implied by the stationary transition-matrix generator for the four-variable VAR(1) simulation condition. The moments of the set points are analytical. The moments of the transition matrices are calculated by Monte Carlo integration over the stationary region for each nonzero heterogeneity condition.

Usage

```
data(populationk4)
```

Format

A list with two elements:

conditions Named list indexed by heterogeneity. Each element contains the calibrated means and covariance matrices, a named population parameter vector, and Monte Carlo calibration information for the four-variable model.

calibration Metadata describing the generator, package version, random seed, population size, batch size, stationarity margin, and generation time.

Author(s)

Ivan Jacob Agaloos Pesigan

results *Simulation Results*

Description

Simulation Results

Usage

```
data(results)
```

Format

A data frame with one row per design condition, method, interval type, and parameter:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible replications contributing to the summary.

success_rate Proportion of requested replications that were admissible and contributed to the summary.

parnames Parameter name.

parameter Calibrated population parameter value. For transition parameters under stability rejection, this is the realized truncated population value rather than the nominal untruncated value.

method Estimation method: MetaVAR, BMLVAR with default priors, BMLVAR with user-defined priors, or the uncertainty-uncorrected method.

n Number of persons.

time Number of measurement occasions; `NA` denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

ci Confidence or credible interval method.

est Mean parameter estimate.

se Mean standard error or posterior standard deviation.

z Mean test statistic when available.

p Mean p-value when available.

ll Mean lower interval limit.

ul Mean upper interval limit.

sig Proportion statistically significant when available.

zero_hit Proportion of intervals containing zero.

theta_hit Proportion of intervals containing the calibrated population value.

sq_error Mean squared error.

bias Mean estimate minus the calibrated population value.

rel_bias Relative bias for nonzero population values and `NA` when the population value is zero.

rmse Root mean squared error.

n_bias Number of finite replication-level bias values used to calculate the Monte Carlo standard error of bias.

n_rmse Number of finite replication-level squared errors used to calculate the Monte Carlo standard error of RMSE.

n_coverage Number of finite coverage indicators used to calculate the Monte Carlo standard error of coverage.

n_rejection Number of finite rejection indicators used to calculate the Monte Carlo standard error of the rejection rate.

mcse_bias Monte Carlo standard error of bias.

mcse_rmse Delta-method Monte Carlo standard error of RMSE, obtained from the Monte Carlo standard error of mean squared error.

coverage Coverage probability.

mcse_coverage Binomial Monte Carlo standard error of coverage.

rejection_rate Proportion of intervals excluding zero.

mcse_rejection_rate Binomial Monte Carlo standard error of the rejection rate.

power Rejection rate for nonzero population values and `NA` for zero population values.

mcse_power Monte Carlo standard error of power for nonzero population values and `NA` for zero population values.

type1_error Rejection rate for zero population values and `NA` for nonzero population values.

mcse_type1_error Monte Carlo standard error of the Type I error rate for zero population values and `NA` for nonzero population values.

par_idx Parameter display index.

Author(s)

Ivan Jacob Agaloos Pesigan

`resultsk4` *Simulation Results (k = 4)*

Description

Simulation Results (k = 4)

Usage

```
data(resultsk4)
```

Format

A data frame with one row per design condition, method, interval type, and parameter:

taskid Simulation task ID.

replications Number of replications.

replications_used Number of admissible replications contributing to the summary.

success_rate Proportion of requested replications that were admissible and contributed to the summary.

parnames Parameter name.

parameter Calibrated population parameter value. For transition parameters under stability rejection, this is the realized truncated population value rather than the nominal untruncated value.

method Estimation method: MetaVAR, BMLVAR with default priors, BMLVAR with user-defined priors, or the uncertainty-uncorrected method.

n Number of persons.

time Number of measurement occasions; NA denotes the unbalanced condition.

heterogeneity Transition-parameter heterogeneity multiplier: 0, 1, or 2.

ci Confidence or credible interval method.

est Mean parameter estimate.

se Mean standard error or posterior standard deviation.

z Mean test statistic when available.

p Mean p-value when available.

ll Mean lower interval limit.

ul Mean upper interval limit.

sig Proportion statistically significant when available.

zero_hit Proportion of intervals containing zero.

theta_hit Proportion of intervals containing the calibrated population value.

sq_error Mean squared error.

bias Mean estimate minus the calibrated population value.

rel_bias Relative bias for nonzero population values and NA when the population value is zero.

rmse Root mean squared error.

n_bias Number of finite replication-level bias values used to calculate the Monte Carlo standard error of bias.

n_rmse Number of finite replication-level squared errors used to calculate the Monte Carlo standard error of RMSE.

n_coverage Number of finite coverage indicators used to calculate the Monte Carlo standard error of coverage.

n_rejection Number of finite rejection indicators used to calculate the Monte Carlo standard error of the rejection rate.

mcse_bias Monte Carlo standard error of bias.

mcse_rmse Delta-method Monte Carlo standard error of RMSE, obtained from the Monte Carlo standard error of mean squared error.

coverage Coverage probability.

mcse_coverage Binomial Monte Carlo standard error of coverage.

rejection_rate Proportion of intervals excluding zero.

mcse_rejection_rate Binomial Monte Carlo standard error of the rejection rate.

power Rejection rate for nonzero population values and NA for zero population values.

mcse_power Monte Carlo standard error of power for nonzero population values and NA for zero population values.

type1_error Rejection rate for zero population values and NA for nonzero population values.

mcse_type1_error Monte Carlo standard error of the Type I error rate for zero population values and NA for nonzero population values.

par_idx Parameter display index.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sim` *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.
`data`	Logical. Simulate data.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVAR` *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVARK4` *Simulation Replication - FitDTVARK4*

Description

Simulation Replication - FitDTVARK4

Usage

```
SimFitDTVARK4(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMetaVAR` *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVARK4 *Simulation Replication - FitMetaVARK4*

Description

Simulation Replication - FitMetaVARK4

Usage

```
SimFitMetaVARK4(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by `.SimSuffix()`.
overwrite	Logical. Overwrite existing output in `output_folder`.
integrity	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus	*Simulation Replication - FitMplus*

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusDiagnostics *Simulation Replication - FitMplus Diagnostics*

Description

Simulation Replication - FitMplus Diagnostics

Usage

```
SimFitMplusDiagnostics(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusK4 *Simulation Replication - FitMplusK4*

Description

Simulation Replication - FitMplusK4

Usage

```
SimFitMplusK4(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusK4Diagnostics

Simulation Replication - FitMplusK4 Diagnostics

Description

Simulation Replication - FitMplusK4 Diagnostics

Usage

```
SimFitMplusK4Diagnostics(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Simk4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusK4PriorDiag *Simulation Replication - FitMplusK4 (User Defined Priors) Diagnostics*

Description

Simulation Replication - FitMplusK4 (User Defined Priors) Diagnostics

Usage

```
SimFitMplusK4PriorDiag(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusK4Priors — *Simulation Replication - FitMplusK4 (User Defined Priors)*

Description

Simulation Replication - FitMplusK4 (User Defined Priors)

Usage

```
SimFitMplusK4Priors(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMplusPriors` *Simulation Replication - FitMplus (User Defined Priors)*

Description

Simulation Replication - FitMplus (User Defined Priors)

Usage

```
SimFitMplusPriors(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplusPriorsDiagnostics
Simulation Replication - FitMplus Diagnostics (User Defined Priors)

Description

Simulation Replication - FitMplus Diagnostics (User Defined Priors)

Usage

```
SimFitMplusPriorsDiagnostics(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitNaive *Simulation Replication - FitNaive*

Description

Simulation Replication - FitNaive

Usage

```
SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by `.SimSuffix()`.
overwrite	Logical. Overwrite existing output in `output_folder`.
integrity	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", "fit-mplus", and "fit-naive".
output_folder	Character string. Output folder.
suffix	Character string. Simulation file suffix returned by .SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Simulation file suffix returned by .SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimGenDataK4` *Simulation Replication - GenDataK4*

Description

Simulation Replication - GenDataK4

Usage

```
SimGenDataK4(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Simulation file suffix returned by `.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `SimK4` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimK4 *Simulation Replication*

Description

Simulation Replication

Usage

```
SimK4(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.
`data`	Logical. Simulate data.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimProj` *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`simulation_diagnostics` *Simulation Diagnostics*

Description

Simulation diagnostics generated from the machine-readable replication status manifests. The objects contain convergence/admissibility counts, boundary or near-zero heterogeneity diagnostics, and runtime summaries.

Usage

```
data(simulation_diagnostics)
```

Format

A list with components `thresholds`, `status`, `status_summary`, `boundary_parameter_replications`, `boundary_parameter_summary`, `boundary_replication_diagnostics`, `boundary_replication_summary`, `runtime_replications`, and `runtime_summary`.

simulation_diagnostics_k4 *Four-Variable Simulation Diagnostics*

Description

Diagnostics for the targeted four-variable feasibility condition.

Usage

```
data(simulation_diagnostics_k4)
```

Format

A list with the same components as simulation_diagnostics.

Sum *Summary*

Description

Summary

Usage

```
Sum(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar_normal,
  metavar_robust,
  mplus,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.

`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`naive`	Logical. If `naive = TRUE`, summarize naive estimates.
`metavar_normal`	Logical. If `metavar_normal = TRUE`, summarize metaVAR model using normal confidence intervals.
`metavar_robust`	Logical. If `metavar_robust = TRUE`, summarize metaVAR model using robust (sandwich) confidence intervals.
`mplus`	Logical. If `mplus = TRUE`, summarize the DSEM model.
`ncores`	Positive integer. Number of cores to use.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumDiagnosticsK4` *Summarize Four-Variable Simulation Diagnostics*

Description

Summarizes the machine-readable status manifests, boundary or near-zero heterogeneity diagnostics, and runtime for a simulation task.

Usage

```
SumDiagnosticsK4(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar,
  mplus,
  variance_tol = 1e-06,
  eigen_tol = 1e-08
)

SumDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
```

```
  integrity,
  naive,
  metavar,
  mplus,
  variance_tol = 1e-06,
  eigen_tol = 1e-08
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
variance_tol	Numeric scalar. Variance estimates no greater than this value are classified as boundary estimates for ML-based methods and as near-zero estimates for Bayesian DSEM.
eigen_tol	Numeric scalar. Estimated heterogeneity covariance matrices with minimum eigenvalues no greater than this value are classified as near singular.

Value

The output is saved as an external file in output_folder and is returned invisibly.

SumFitMetaVARK4Normal *Summary: Normal Theory (FitMetaVARK4)*

Description

Summary: Normal Theory (FitMetaVARK4)

Usage

```
SumFitMetaVARK4Normal(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via `SumK4()`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMetaVARK4Robust` *Summary: Robust (FitMetaVARK4)*

Description

Summary: Robust (FitMetaVARK4)

Usage

```
SumFitMetaVARK4Robust(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via SumK4().

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARRobust *Summary: Robust (FitMetaVAR)*

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus *Summary (FitMplus)*

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplusDiagnostics *Summary (FitMplus Diagnostics)*

Description

Summary (FitMplus Diagnostics)

Usage

```
SumFitMplusDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplusK4` *Summary (FitMplusK4)*

Description

Summary (FitMplusK4)

Usage

```
SumFitMplusK4(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via `SumK4()`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplusK4Diagnostics` *Summary (FitMplusK4 Diagnostics)*

Description

Summary (FitMplusK4 Diagnostics)

Usage

```
SumFitMplusK4Diagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity` = `TRUE`, check for the output file integrity when `overwrite` = `FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via `SumK4()`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplusK4Priors` *Summary (FitMplusK4 with User-Defined Priors)*

Description

Summary (FitMplusK4 with User-Defined Priors)

Usage

```
SumFitMplusK4Priors(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity` = `TRUE`, check for the output file integrity when `overwrite` = `FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via `SumK4()`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplusK4PriorsDiagnostics`
Summary (FitMplusK4 User-Prior Diagnostics)

Description

Summary (FitMplusK4 User-Prior Diagnostics)

Usage

```
SumFitMplusK4PriorsDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via `SumK4()`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplusPriors *Summary (FitMplus (User Defined Priors))*

Description

Summary (FitMplus (User Defined Priors))

Usage

```
SumFitMplusPriors(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplusPriorsDiagnostics *Summary (FitMplus Diagnostics with User Defined Priors)*

Description

Summary (FitMplus Diagnostics with User Defined Priors)

Usage

```
SumFitMplusPriorsDiagnostics(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in `output_folder`.
integrity	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitNaive` *Summary: Naive*

Description

Summary: Naive

Usage

```
SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumK4 *Summary for the Four-Variable Simulation*

Description

Summary for the Four-Variable Simulation

Usage

```
SumK4(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar_normal,
  metavar_robust,
  mplus,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in `output_folder`.
integrity	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
naive	Logical. Retained for consistency with `SimK4()`. The four-variable simulation does not currently fit a naive model.
metavar_normal	Logical. If `TRUE`, summarize the four-variable metaVAR fits using normal confidence intervals.
metavar_robust	Logical. If `TRUE`, summarize the four-variable metaVAR fits using robust confidence intervals.
mplus	Logical. If `TRUE`, summarize the default- and user-prior four-variable DSEM fits and their diagnostics.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as external files in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`TabMplusDiagSensitivity`

Bayesian Diagnostic Failure-Rate Sensitivity Table

Description

Create a simulation-case-level table of Bayesian Mplus diagnostic failure rates. Unlike the overview figure, the function retains one row for every simulation task ID and does not average across simulation cases.

Usage

```
TabMplusDiagSensitivity(
  diagnostics,
  comparison = "difference",
  diagnostic = "Any diagnostic",
  unit = "replication",
  taskid = NULL,
  target = NULL,
```

```
  rhat_threshold = 1.01,
  ess_threshold = 400,
  mcse_threshold = 0.05,
  relative_mcse = TRUE,
  output = "data.frame",
  scale = "percent",
  digits = 0L
)
```

Arguments

diagnostics	An all-task diagnostics object containing a parameters data frame, or the parameter-level data frame itself.
comparison	Character string. "difference" returns paired user-prior minus default-prior failure-rate differences. "levels" returns the two prior-specific failure rates. "all" returns the default rate, user-prior rate, and paired difference.
diagnostic	One or more diagnostic panels. Available values are "Any diagnostic", "R-hat", "Bulk ESS", "Tail ESS", and "Relative MCSE" when relative_mcse = TRUE.
unit	Character string controlling the unit classified as failed. "replication" classifies a replication as failed when any parameter in the selected block fails. "parameter" averages failures over parameter-replication pairs.
taskid	Optional positive integer task IDs to include.
target	Optional parameter blocks to include. Available values are "Innovation", "FE", and "RE".
rhat_threshold	Numeric R-hat threshold.
ess_threshold	Numeric bulk and tail effective sample size threshold.
mcse_threshold	Numeric Monte Carlo standard error threshold.
relative_mcse	Logical. If TRUE, Monte Carlo standard errors are divided by the posterior standard deviation before classifying failures.
output	Character string. "data.frame" returns a wide data frame with simulation-design columns, "matrix" returns only the numeric case-by-summary matrix, and "long" returns one row per task, block, diagnostic, and prior comparison.
scale	Character string. "proportion" returns values on the 0-to-1 scale. "percent" multiplies prior-specific failure rates by 100 and expresses differences in percentage points.
digits	Nonnegative integer used to round the returned values.

Details

Failure rates are computed separately within each task ID. For comparison = "difference", replications are paired before calculating the user-prior minus default-prior difference. Negative differences favor the user-specified priors.

The default unit = "replication" answers the question: within a given simulation case, what percentage of replications had at least one failed parameter in the selected parameter block?

Value

A data frame or numeric matrix. For output = "matrix", simulation design information is stored in the "case_data" attribute.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
## Not run:
data(diagnostics, package = "manMetaVAR")

# Compact sensitivity matrix: one row per simulation case
TabMplusDiagSensitivity(
  diagnostics = diagnostics,
  comparison = "difference",
  output = "matrix"
)

# Default, user-prior, and difference columns
TabMplusDiagSensitivity(
  diagnostics = diagnostics,
  comparison = "all",
  output = "data.frame"
)

# Diagnostic-specific table
TabMplusDiagSensitivity(
  diagnostics = diagnostics,
  comparison = "difference",
  diagnostic = c("R-hat", "Bulk ESS", "Tail ESS", "Relative MCSE")
)

## End(Not run)
```

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